



WFH Burnout Predictor

Predicting work-from-home burnout score from daily behavioral signals.

A CRISP-DM regression and classification project, evaluated on employees unseen during training.

Can daily behavior predict burnout before it peaks?

Predicting `burnout_score` from work-from-home indicators such as workload, screen time, meetings, breaks, after-hours work, sleep and task completion; evaluated on employees not seen during training.

Why this approach

- Grouped train / test split: no employee appears on both sides.
- Baselines first: dummy models set the bar before complex ones.
- Explicit limits: claims are scoped to what the data supports.



Main task

Regression on `burnout_score`: the continuous burnout level for each employee.



Secondary task

Classification of `burnout_risk` (Low / Medium / High), used mainly for interpretation.



Focus

The dataset is technically simple, so the project emphasizes honest data and result understanding.



Business
Understanding



Data
Understanding



Data
Preparation



Modeling



Evaluation



Deployment

A clean, repeated-measures dataset

Each of the 180 employees contributes 10 daily records, so the identifier is useful for evaluation design (grouping train and test) but not as a predictive feature.



1,800

observations



180

employees (users)



10

records per user



0

missing values

Predictors — 8 daily behavioral signals

day_type

work_hours

screen_time_hours

meetings_count

breaks_taken

after_hours_work

sleep_hours

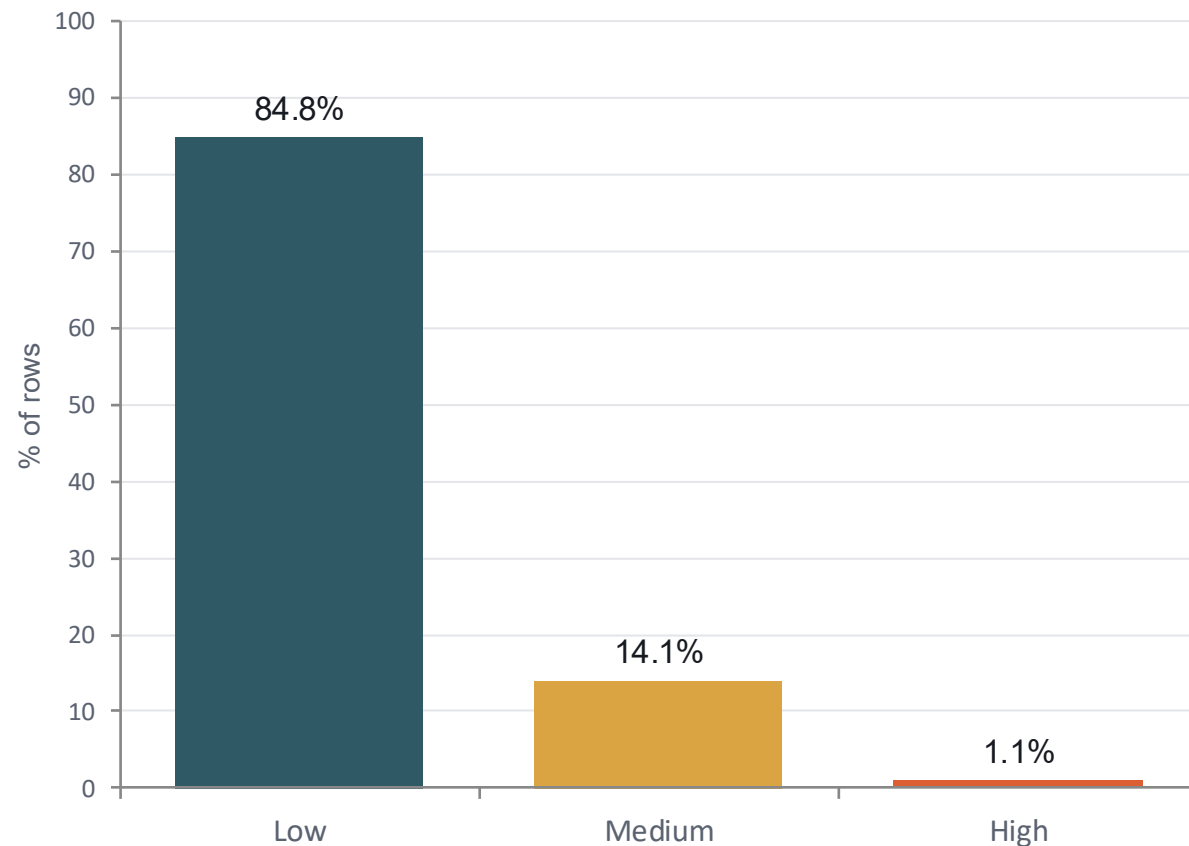
task_completion_rate

Targets

- **burnout_score** continuous · regression target
- **burnout_risk** Low / Medium / High · classification target

Technical quality is high: no missing values, no duplicates, valid ranges. But external representativeness beyond this public sample remains unverified.

burnout_risk leans heavily toward Low



Why this matters

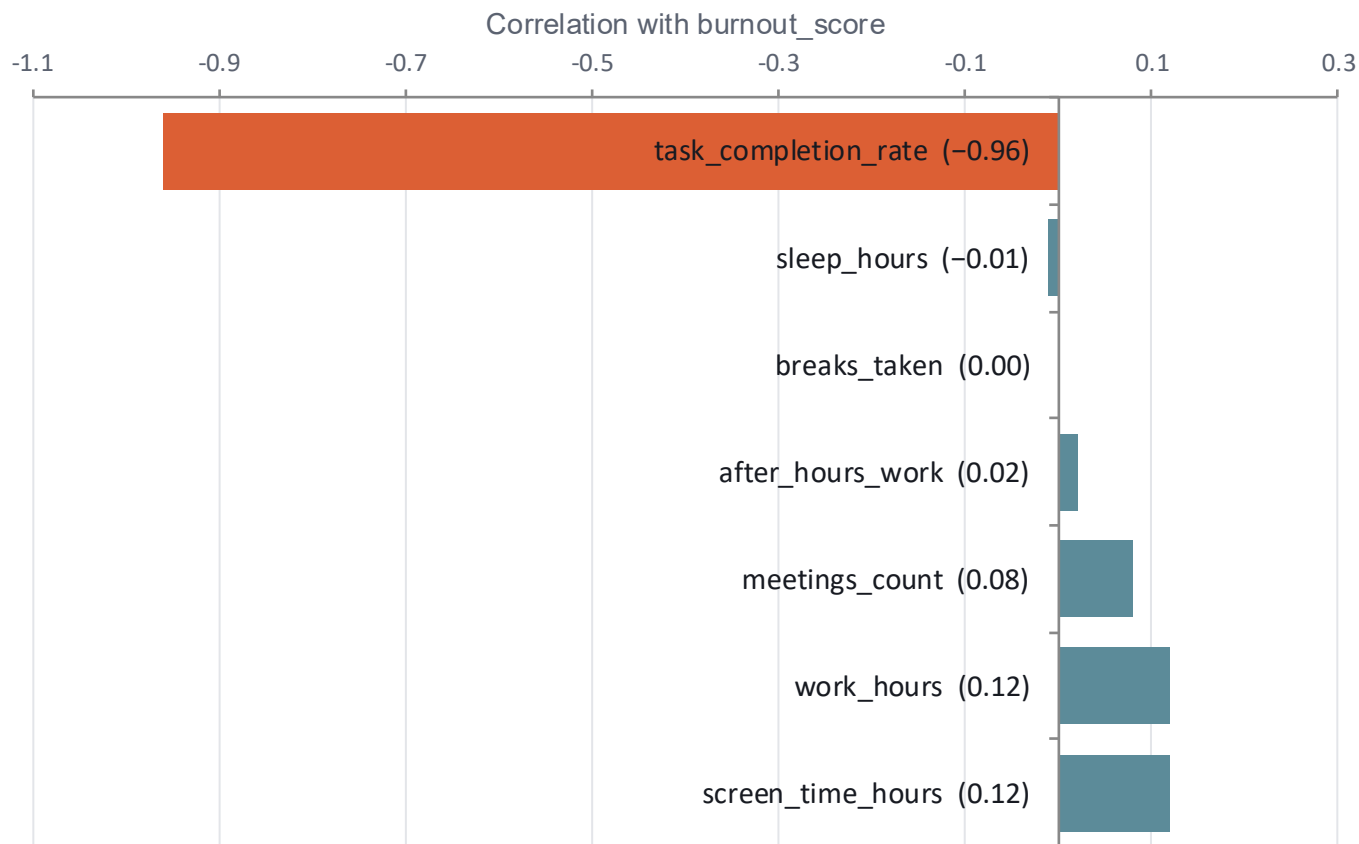
- Low-risk rows dominate the dataset — 1,527 of 1,800 rows.
- High risk appears in only 20 records — about 1.1%.
- A classifier can look accurate while missing the rarest class entirely.
- The continuous burnout_score is therefore treated as the main target.

Note: burnout_risk is a deterministic bucketisation of burnout_score (Low ≤ 69.92 · Medium 70.01–109.99 · High ≥ 110.22), verified programmatically. A good regressor can therefore be thresholded into a near-perfect classifier; the classification task adds interpretability, not new predictive signal.

20 rows

out of 1,800 carry the High-risk label

One feature carries almost all the signal



-0.96

Correlation between task_completion_rate and burnout_score.

This is useful for prediction, but it is also the main warning sign of the project: the feature may sit very close to the logic that generated the target in the first place.

All other predictors stay within ± 0.12 — a much weaker, more plausible behavioral signal.
A univariate Ridge fitted on task_completion_rate alone reaches test $R^2 = 0.934$ — nearly identical to the full thirteen-feature model (0.936). The remaining predictors contribute almost no incremental signal.

The split is designed around people, not rows

If the same employee appeared in both train and test, the model could memorize personal baselines instead of learning generalizable patterns.



Excluded

user_id is never used as a predictor — it has no generalizable meaning.

1,440 / 360

train / test rows



Grouped split

Records from the same employee stay entirely in train or entirely in test.

144 / 36

train / test users



Pipeline

Encoding, scaling and model fitting all run inside scikit-learn pipelines.

0

overlapping users

Engineered features: interpretable workload-pressure signals

work_intensity

screen_work_ratio

meeting_density

sleep_deficit

overwork_flag

Baselines first, then increasingly complex models



Baseline

Dummy models define the minimum acceptable result.



Linear

Ridge checks whether the signal is mostly linear.



Non-linear

Random Forest and Gradient Boosting capture interactions.



Validation

GroupKFold keeps user-level grouping during model selection.

Cross-validation comparison (training set, GroupKFold)

Model	CV MAE	CV R ²
Random Forest	4.97	0.923
Ridge	5.06	0.916
Hist Gradient Boosting	5.19	0.917
KNN	9.03	0.744
Dummy (median)	18.46	-0.046

Random Forest wins on cross-validation and is selected for tuning.

All meaningful models clear the dummy baseline

Model	MAE	RMSE	R ²
Ridge	4.79	6.31	0.934
Random Forest	4.85	6.41	0.932
Hist Gradient Boosting	5.14	6.73	0.925
KNN	9.26	11.65	0.776
Dummy (median)	20.35	25.80	-0.099

Test-set results, before hyperparameter tuning.

Ridge shows the lowest raw test MAE (4.79); Random Forest was selected via cross-validated MAE (4.97 vs. 5.06 for Ridge) — a more reliable criterion than a single 360-row draw.

TUNED MODEL

Random Forest Regressor

4.88

Test MAE

6.21

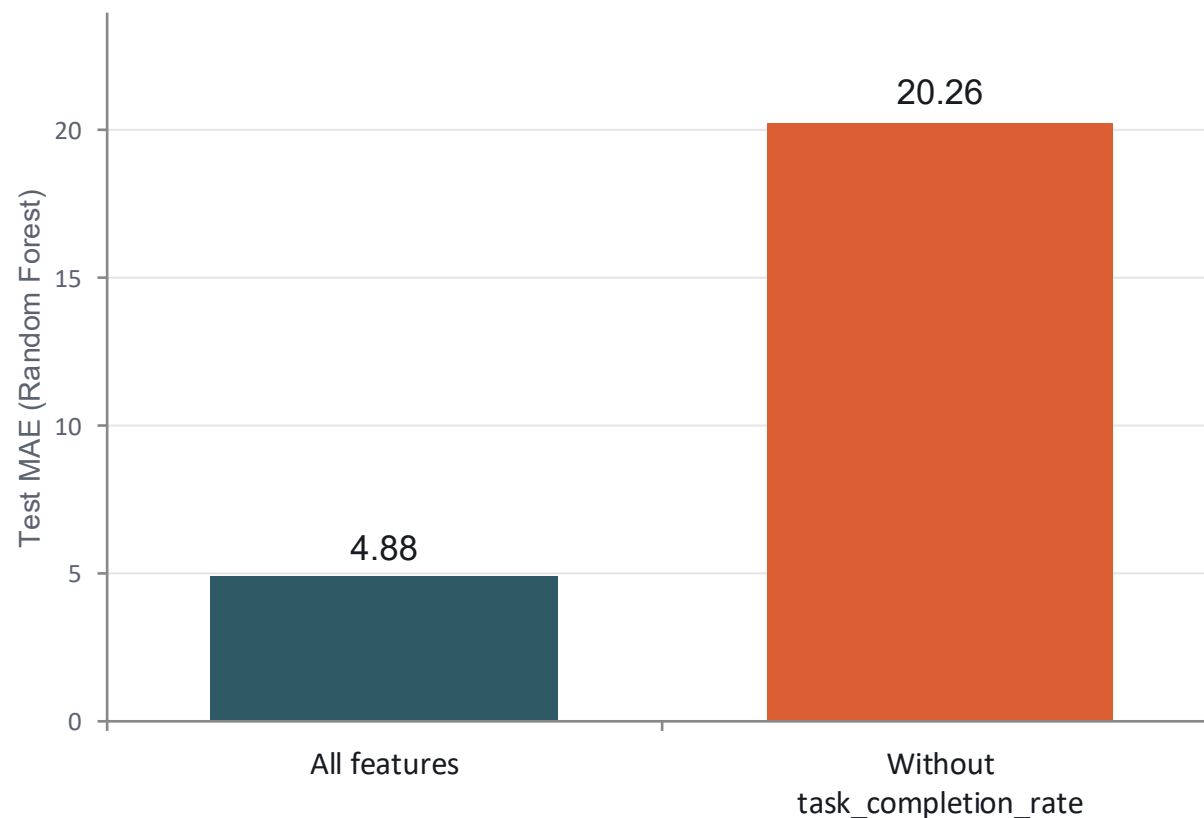
Test RMSE

0.936

Test R²

200 trees · max depth 4 · min leaf 8, selected via
RandomizedSearchCV on training-set GroupKFold.

Remove one feature, and performance collapses



task_completion_rate alone carries almost all of the predictive power.

Dropping it pushes Random Forest MAE from 4.88 to 20.26, within noise of the dummy median baseline (≈ 20.35).

This does not invalidate the pipeline.

It narrows the claim: the project demonstrates predictive modeling and is explicit about exactly where its reliability comes from.

Risk labels help, but the rarest class stays hard to catch

0.803

Balanced accuracy

0.646

Macro F1

Balanced Logistic Regression

Best classifier

Risk class	Precision	Recall	F1	Support
Low	0.99	0.93	0.96	294
Medium	0.68	0.73	0.70	62
High	0.17	0.75	0.27	4

Precision on High is low (because of many false alarms) but **recall** of **0.75** means **most true high-risk test cases are caught**. That recall is computed on just 4 examples, so should be treated as a direction, not a precise rate.

A cautious decision-support tool, not an automatic verdict



Interpretability

Permutation importance confirms `task_completion_rate` dominates the prediction, but it should not be read causally.



Fairness checks

The dataset has no classic sensitive attributes. Errors and high-risk recall are still checked across `day_type`, `after_hours_work` and work-hour bands.



Deployment plan

Save the full pipeline, not just the model.
Monitor feature and performance drift.
Repeat leakage and fairness audits over time.



External validation on real organizational data is required before any operational use — this remains a research-stage pipeline.



What this project actually demonstrates

01

Data understanding drives the project

Clean but simple data, repeated users, and a rare High-risk class shape every later choice.

02

Evaluation is conservative

Users are grouped end-to-end, so no employee leaks between training and test.

03

The regression result is strong

Tuned Random Forest reaches MAE 4.88 and R^2 0.936 on employees unseen during training.

04

The main limitation is explicit

Performance leans heavily on `task_completion_rate`.

05

Final read

A transparent predictive pipeline but there is not evidence of causal burnout mechanisms.